Sensitivity Is Not Fragility: Shared Visual Responses in Vision-Language-Action Policies

Anonymous CVPR submission

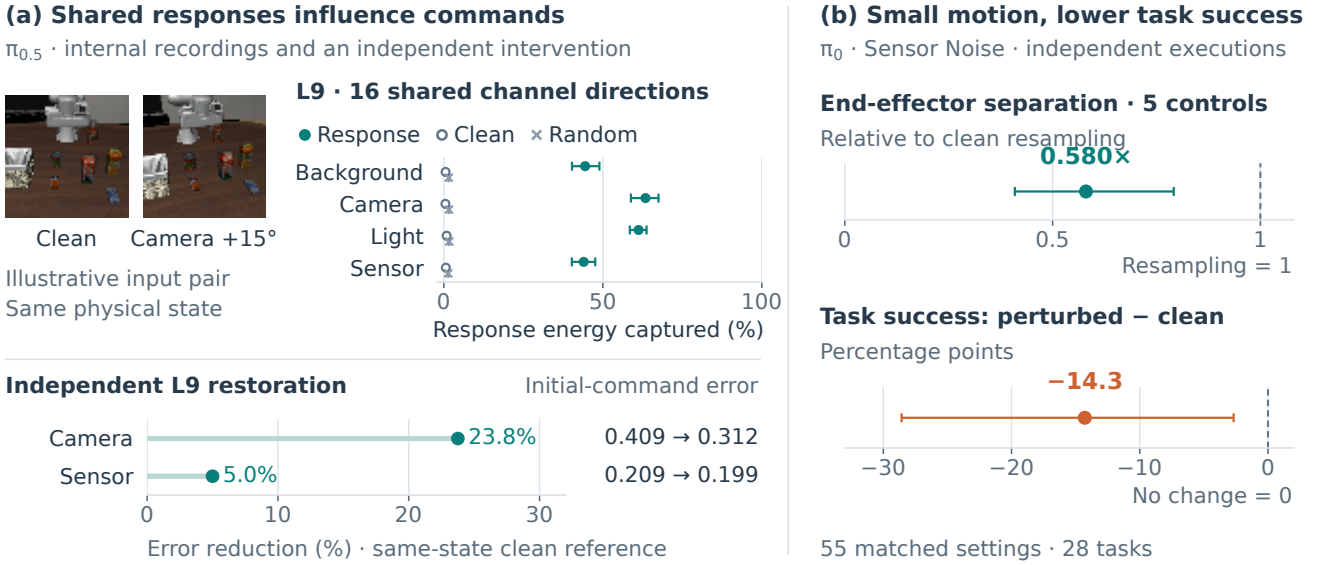


Figure 1. **Shared responses support action correction; small motion accompanies task loss.** (a) Sixteen shared L9 directions retain more centered response energy than equal-rank clean and random directions on held-out tasks and visual families (404 pairs; pointwise 95% task intervals). In a separate experiment, restoring their clean coordinates reduces task-mean initial-command error by the percentages shown (Fig. 4). The example images come from the separate fixed-state camera study (Supplement, Sec. S20). (b) Under Sensor Noise, π_0 loses task success even though early-motion differences are smaller than those between two clean runs. Both measurements use the same 55 settings and 28 tasks; intervals are 95% for motion and four-family-adjusted 98.75% for success.

Abstract

Visual perturbations cause substantial losses in VLA task success, yet action changes alone do not reveal their task consequences. In our experiments, visual conditions that induce markedly different action deviations yield the same success counts, while small early-motion deviations in another setting accompany substantial success losses. These contrasts motivate a closer study of which internal responses recur across visual conditions, whether restoring them can correct action generation, and how action changes relate to robot execution. We investigate these questions in frozen $\pi_{0.5}$ and π_0 on LIBERO and LIBERO-Plus through controlled visual changes, internal interventions, and comparisons of robot behavior. In $\pi_{0.5}$, perturbation-induced feature changes share a low-dimensional structure that extends to held-out tasks and perturbation families. Restoring the shared components using clean observations of the same physical state brings the first five translation commands closer to their clean references; the tested camera and sensor conditions also show higher observed success rates. Finally, independent execution studies show that initial command changes closely track early motion, while small average motion deviations can still accompany substantial success losses. Unchanged overall success can also conceal individual executions switching in opposite directions between success and failure. These findings identify reusable components for studying internal corrections to VLA visual sensitivity, and show why robot motion and individual task outcomes must both inform the evaluation of visual robustness.

1. Introduction

Changes in background, viewpoint, or lighting leave the robot's task unchanged, yet strongly affect VLA behavior. For example, LIBERO-Plus reports that π_0's success rate falls from 94.2% on the original LIBERO benchmark to 13.8% under camera perturbations [2]. However, large action deviations need not prevent task completion. In our $\pi_{0.5}$ lighting study, the mean deviation of the first five translation commands from their unperturbed references differs by a factor of 5.12 between two perturbations, yet both perturbed conditions and the unperturbed condition achieve 77 successes in 80 executions (Supplement, Sec. S10.6). Conversely, in π_0 under sensor noise, we observe substantial success losses even when average early-motion deviations are smaller than variation between unperturbed runs. These contrasts motivate three questions: do different visual perturbations induce shared internal responses, can restoring those responses correct generated actions, and what do action changes reveal about task execution?

Shared responses would offer a concrete way to study visual robustness: different disturbances could be examined through common internal components, and those components could be tested directly during action generation. We therefore learn directions from perturbation-induced feature differences and test their reuse on tasks and perturbation families excluded from fitting. We then restore their clean coordinates using an unperturbed observation of the same physical state. This tests whether the shared structure can be used to correct the policy's own output. We also measure task success to assess how this intervention changes task completion.

We study frozen, LIBERO-finetuned $\pi_{0.5}$ [8] and π_0 [1] on LIBERO [5] and LIBERO-Plus under changes in background, camera viewpoint, lighting, and sensor noise. Separate execution studies with both policies compare action commands, robot motion, and task outcomes with and without perturbations (Table 1). We use *sensitivity* to describe changes in internal features, commands, or motion, and *fragility* to describe reduced task success. Our main findings are:

Shared perturbation responses extend across tasks and visual conditions. In $\pi_{0.5}$, 16 shared channel directions reconstruct perturbation-induced changes on jointly held-out tasks and perturbation families more accurately than equal-rank clean-feature and random controls. Response reconstruction and command prediction with a fixed affine readout favor different sampled layers. These directions provide a common basis for analyzing visual sensitivity across conditions (Fig. 1; Sec. 4).

Restoring shared components corrects initial translation commands. In an independent confirmation study, restoring the shared components using clean observations of the same physical state reduces the clean-reference error of the first five translation commands relative to unedited generation and clean-basis or random-basis restoration. This correction requires no retraining; the tested camera and sensor conditions also show higher observed task success (Fig. 4; Sec. 4).

Early motion tracks action changes, but small deviations can accompany task loss. Translation-command shifts recur across the two recorded noise inputs used to generate actions and are more strongly associated with early motion than with later motion after replanning. Under sensor noise, π_0 loses 14.3 percentage points in task success despite average early-motion deviations below variation between unperturbed runs. For $\pi_{0.5}$, background changes leave overall observed success unchanged, while some previously successful executions fail and some previously unsuccessful executions succeed. Tracking these opposing outcome changes reveals effects on task completion that are hidden by the net success rate (Sec. 5).

2. Related Work

Visual robustness of VLA policies. Robustness studies examine whether VLA policies maintain task performance when their observations or environments change. LIBERO-Plus [2] systematically evaluates variations in background, viewpoint, illumination, and sensor noise, revealing substantial success losses under changed visual conditions. Alongside these evaluations, robustness methods intervene at different stages: BYOVLA [3] edits task-irrelevant image regions during deployment, while RoVLA [6] encourages consistent predictions under observation and instruction changes during training.

Mechanistic analysis of VLA policies. VLA-Trace [10] traces representations and intervenes on attention to study how multimodal information supports action generation. Related studies examine language grounding through activation patching [12] and relate interventional visual attribution to generalization [13]. Sparse-autoencoder analyses [11] identify reusable motion and semantic features whose manipulation steers robot behavior. COAST [7] constructs activation subspaces from successful and failed rollouts and uses them to improve policy execution.

Evaluating robot behavior beyond task success. Higham et al. [4] characterize the smoothness, efficiency, and gripper behavior of successful trajectories under input perturbations, showing that similar success rates can accompany different execution behavior. Sawada and Kasahara [9] compare matched executions under individual and combined perturbations. Combining perturbations can cause a previously successful execution to fail or a failed execution to succeed; these opposing transitions can cancel in aggregate success rates.

Position of our work. We study the changes induced by visual perturbations: which internal components recur

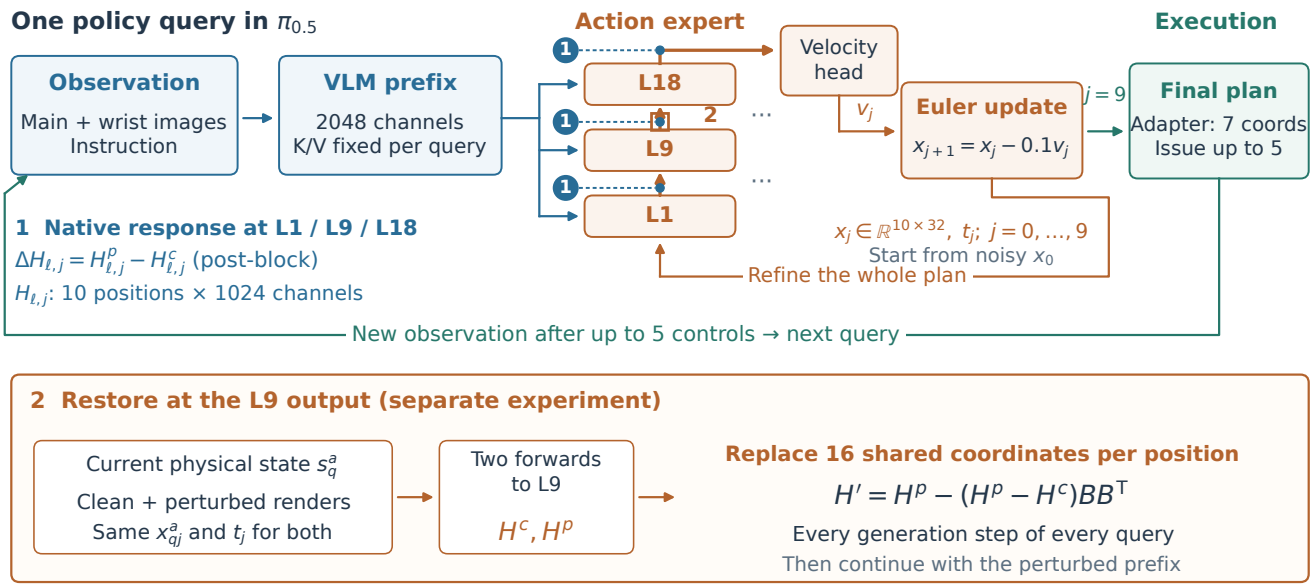


Figure 2. **Native measurement and coordinate restoration within two nested loops.** Within a $\pi_{0.5}$ query, a fixed visual–language prefix conditions ten updates of the entire ten-position plan; up to five controls are then issued before observing again. Dashed blue taps locate native post-block measurements at L1/L9/L18 of the 18-layer action expert. Clean and perturbed recordings follow their own generation paths. In the separate restoration experiment (orange square and inset), two forwards use the intervention run's current physical state and the same generation state and time. The rank-16 edit is written after L9 at every generation step of every query; subsequent layers retain the perturbed prefix. B contains 16 shared directions in the 1024-dimensional channel space.

across tasks and perturbation families, and whether restoring their clean coordinates corrects the resulting action deviations. Joint task/family holdouts test reuse beyond the fitting conditions; restoration during the policy's own generation tests effects on action commands. Independent executions compare motion against variation between clean runs and track lost and gained successes, linking perturbation sensitivity to task consequences. Complementary attention analysis appears in Supplement, Sec. S20.

3. Analysis Framework

We study frozen, LIBERO-finetuned $\pi_{0.5}$ [8] and π_0 [1] under four LIBERO-Plus visual perturbation families: Background, Camera, Light, and Sensor Noise. Internal recordings connect feature responses to generated commands; restoration experiments test whether changing those features corrects commands. Independently collected executions compare robot motion and task outcomes. Each clean–perturbed comparison keeps the task unchanged and matches the initial physical state and action-generation noise. Table 1 distinguishes these studies and their support.

3.1. From an Observation to Issued Commands

At each policy call (*query*), the vision-language network encodes the main-camera image, wrist-camera image, and task instruction into context tokens, called the *prefix* (Fig. 2).

Study	Evaluation support
Internal recordings ($\pi_{0.5}$)	536 pairs / 40 tasks; primary: 404 pairs / 28 tasks (Secs. 4.1–4.2)
Paired executions (both policies)	Per policy: 268 settings / 40 tasks; primary: 202 settings / 28 tasks (Sec. 5)
Native restoration ($\pi_{0.5}$)	Independent confirmation: 31 tasks, 96 Camera and 120 Sensor pairs (Sec. 4.3)

Table 1. **Three studies link internal responses, corrections, and execution.** Internal recordings and executions use separate episodes from a common visual-setting plan. The primary internal panel has two noise-matched pairs per setting; restoration uses independent confirmation tasks.

The evaluated π_0 configuration additionally receives a robot-state token; for the $\pi_{0.5}$ runs analyzed here, joint values are not appended to the instruction text (Supplement, Secs. S15 and S11.2). We denote the fixed conditioning inputs for a policy call by C. The *action expert* uses these inputs to generate an action plan: a sequence of future actions.

Starting from a noise sample x_0, the action expert refines the entire plan through ten Euler updates. Let x_j denote the full action plan at generation step j:

$$x_{j+1} = x_j + \Delta t_j F_\theta(C, x_j, t_j), \quad j = 0, \ldots, 9. \quad (1)$$

Here t_j is generation time, F_θ predicts the plan's update velocity, $t_0 = 1$, and $\Delta t_j = -1/10$. Plans contain ten actions for $\pi_{0.5}$ and fifty for π_0, each with 32 model coordinates. A *generation step* updates the whole plan before execution; a *plan position* identifies one future action.

Once generation finishes, an adapter converts each action into seven simulator-control coordinates for translation, rotation, and the gripper. The robot executes up to the first five commands, then obtains a new observation and generates another plan. We count an *issued control* when its simulator call returns. The first five controls therefore provide an early execution window driven by the initial plan, before feedback from subsequent observations enters.

To examine how visual perturbations affect this generation process, we record $\pi_{0.5}$'s hidden states immediately after action-expert blocks at layers $\ell \in \{1, 9, 18\}$. At layer ℓ and generation step j, let $H^c_{\ell,j}$ and $H^p_{\ell,j}$ denote the states under clean and perturbed visual conditions. We define the perturbation *response* as their difference:

$$\Delta H_{\ell,j} = H^p_{\ell,j} - H^c_{\ell,j}. \tag{2}$$

For $\pi_{0.5}$, $\Delta H_{\ell,j} \in \mathbb{R}^{10\times 1024}$ contains one row per planned action and 1024 feature channels per row. The branches share the initial noise but update their plans separately, so later responses include differences between the evolving plans.

3.2. Paired Studies and Their Comparisons

A *setting* is one task under a specific visual variant. Internal recordings and execution studies follow a common plan of 268 settings across 40 tasks, using separate episodes. Primary comparisons use 202 LIBERO-Plus settings with benchmark severity labels across the 28 tasks shared by all four perturbation families. Two noise-matched clean–perturbed pairs per setting give 404 internal pairs, each linking features and commands from the same episodes. Restoration uses an independent confirmation set (Table 1).

For each model and setting, the execution study compares clean and perturbed runs under two policy-noise inputs, giving (R_0, T_0) and (R_1, T_1). The two clean runs, R_0 and R_1, measure variation from changing the noise without a visual perturbation: the *clean-resampling reference*. A same-noise repeat R'_0 checks repeatability. All five runs start from a matched state on the same worker and then follow their own observations. Motion is compared after equal numbers of issued controls.

We average execution results over noise inputs within settings, settings within tasks, and tasks with equal weight. Sampling intervals resample whole tasks, retaining their settings and paired runs; they are conditional on the recorded noise inputs and initial states. For missing outcomes, we retain planned denominators and bound results over all feasible assignments of success or failure. These missing-outcome bounds are distinct from sampling intervals (Supplement, Secs. S1, S4, and S6).

4. Shared Response Directions and Effects on Action Commands

Visual perturbations induce shared internal changes in $\pi_{0.5}$ that extend to held-out tasks and perturbation families. The shared directions reconstruct these responses more accurately than equal-rank clean or random directions, and restoring their clean coordinates during the policy's own generation corrects initial action commands. The sampled layers also serve different analysis goals: L1 retains the largest response-energy fraction, whereas L18 best predicts command changes with the fixed linear readout. We establish these findings through separate tests of response reconstruction, action prediction, and internal restoration. Together, they provide reusable components for studying action correction without retraining and a basis for choosing layers according to the question being asked.

4.1. Shared Responses across Tasks and Visual Families

Visual perturbations across tasks produce shared patterns of change inside the action expert of $\pi_{0.5}$. We characterize each response as the difference between perturbed and clean (unperturbed) hidden states. A shared direction describes how multiple feature channels change together. To test whether these patterns extend beyond the conditions used for fitting, we jointly hold out the test task group and its perturbation family; tasks sharing clean references stay in the same group. We fit 16 orthonormal directions separately at each sampled layer (L1, L9, and L18) using principal component analysis (PCA), with the policy frozen.

When the policy first generates an action plan, we record hidden states at the first generation step. For each of the ten planned action positions i, let ΔH_i be the observed response, μ_i its training mean, and $B \in \mathbb{R}^{1024\times 16}$ the learned directions for that layer and holdout:

$$\widehat{\Delta H}_i = \mu_i + (\Delta H_i - \mu_i) B B^\top. \tag{3}$$

The projection retains the response component along these directions. We measure its share of the squared magnitude remaining after subtracting μ_i, termed retained response energy. All ten positions share B, with 16 coordinates per position computed from its observed test response. Directions learned from position-centered clean states test whether ordinary clean-state variation provides an equally effective basis. Random directions test how much energy is retained without learning the response structure. All three bases use the same rank and reconstruct the same centered response.

Across 404 matched pairs covering 28 tasks, shared response directions retain 36.5–83.6% of response energy,

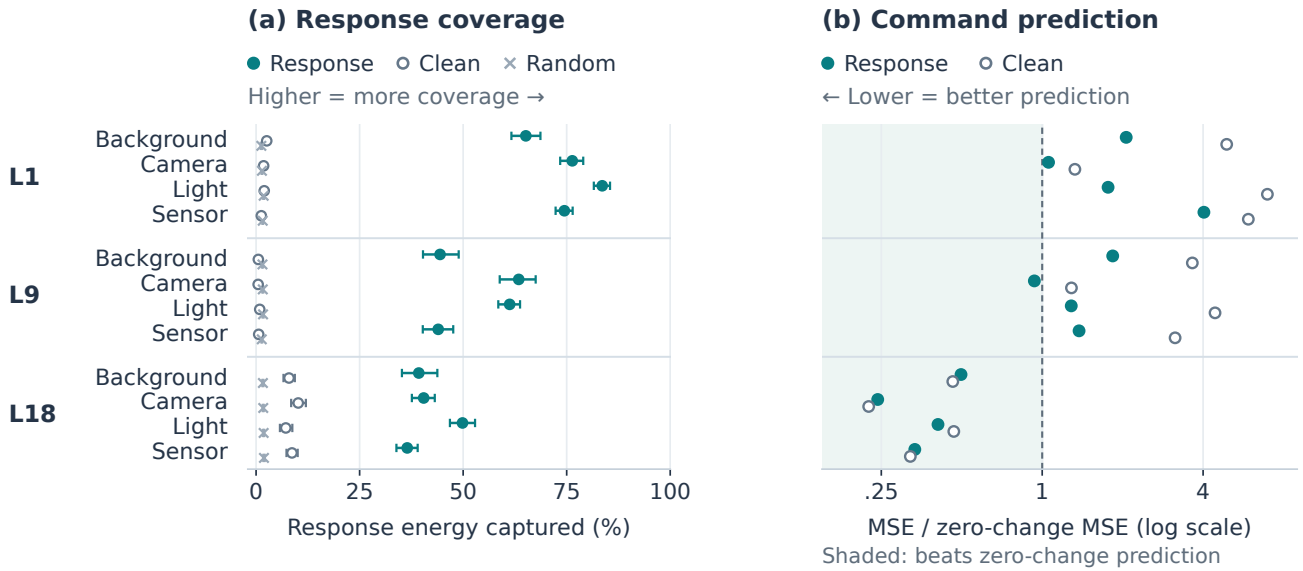


Figure 3. **Response coverage and fixed affine command prediction favor different layers.** Internal recordings: the same 404 pairs on 28 common tasks, with joint task/family holdout. (a) Initial-step centered response energy captured by 16 shared channel directions, with pointwise 95% task intervals. Each plan position uses 16 observed-response coefficients. (b) Affine translation-command prediction error relative to predicting zero change; lower values are better, and shaded values below 1 beat that baseline. Response and clean-basis readouts are point estimates on a log axis. L1 captures the most response energy, while L18 gives the lowest affine prediction errors. Paired basis-comparison intervals and the complete panels appear in Supplement, Sec. S18.

compared with 0.5–10.2% for clean directions and 1.3–1.9% for random directions (Fig. 3a; means with equal task weighting for each layer and family). In all four perturbation families, mean retention is highest at L1, intermediate at L9, and lowest at L18. Thus, directions learned from response differences capture shared variation that equally sized clean-state and random bases miss. The advantage persists when varying the number of directions and comparing against training-mean templates (Supplement, Sec. S3).

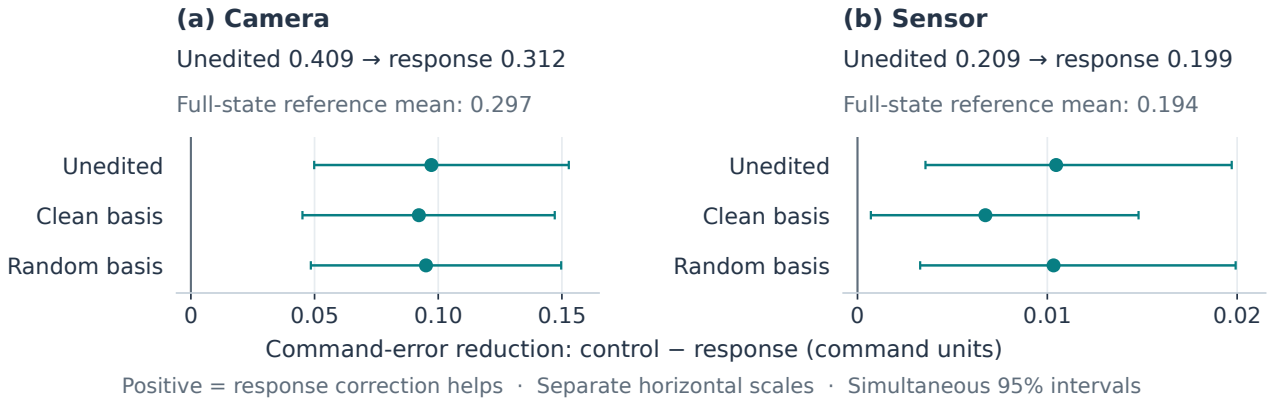
The reconstruction advantage also appears in a separate π_0 analysis, where retention peaks at L9 (Supplement, Sec. S15). An independent LIBERO study further shows that directions learned from earlier observations retain their reconstruction advantage at later observations of held-out tasks under the same visual conditions (Supplement, Sec. S8.1). These results establish reusable response components, with different layer profiles in the two policies. Returning to $\pi_{0.5}$, we next ask whether the layer that retains the largest share of the response also gives the most accurate prediction of action changes.

4.2. Response Reconstruction and Action Prediction Favor Different Layers

In $\pi_{0.5}$, response reconstruction and action prediction with a fixed affine readout favor different sampled layers. We compare them on the same 404 paired cases and task/family holdouts used in Sec. 4.1. An affine ridge readout—a regularized linear mapping with an offset—uses response coordinates normalized using training data to predict changes in the first five translation commands issued at the initial planning query, a 15-dimensional target. We divide its task-averaged mean squared error (MSE) by the task-averaged error of predicting no change from the clean commands. A ratio below 1 therefore means more accurate prediction than this no-change baseline (Fig. 3b).

In every perturbation family, average response retention decreases from L1 through L9 to L18, while average action prediction error also decreases. For Light, the three layers retain 83.6%, 61.2%, and 49.8% of response energy, respectively, whereas their normalized prediction errors are 1.764, 1.284, and 0.407. The difference between the best reconstruction and prediction layers also holds within most tasks: L1 reconstructs best and L18 predicts best in 24 of 28 Background tasks, 26 of 28 Camera tasks, and 27 of 28 tasks for each of Light and Sensor. These comparisons reuse the same 28 tasks across families.

Against the no-change baseline, the response-basis readout reduces mean prediction error in all four families at L18 and in Camera at L9; L9 also improves on L1 in every family. We separately test whether learning directions from perturbation responses improves prediction over directions learned



	Task success (%)			Paired gain (percentage points)
Family	Clean	Perturbed	Response restoration	Gain [simultaneous 95% interval]
Camera	98.15	73.15	75.00	+1.85 [−8.33, 11.11]
Sensor	97.50	83.33	86.67	+3.33 [−1.67, 9.17]

Figure 4. **Restoring shared response coordinates corrects initial commands, with task outcomes measured in the same study.** The plots measure error in the initial five issued translation commands relative to paired native-clean commands. Points show control minus response-restoration error: positive values favor restoration, and all six simultaneous 95% command intervals are positive. For example, the Camera unedited contrast is $0.409 - 0.312 \approx 0.097$. Headers compare mean errors for unedited, response and full-state restoration. The table reports the primary task-success endpoint and gains over the unedited perturbed policy, with simultaneous 95% intervals over six success comparisons. Independent all-step L9 confirmation uses 96 Camera pairs on 27 tasks and 120 Sensor pairs on 30 tasks. The plots have different horizontal scales; each basis restores its own coordinate differences, giving different write magnitudes. Complete command and success controls appear in Supplement, Sec. S13.

from clean states, using the same affine readout specification and command-change targets. Response coordinates outperform clean coordinates at L1 and L9. At L18, both bases outperform no change, while the paired relative error-reduction intervals for response over clean coordinates span zero (Supplement, Sec. S3.4, Table S6; Sec. S14, Table S36). Additional readouts show that these basis rankings depend on the fitted estimator (Supplement, Sec. S14).

The shared directions therefore serve distinct analysis goals, with different best layers in this fixed-readout comparison. An external predictor tests what can be read from the response coordinates; restoring them inside the policy tests what they can change. We next use the latter test to examine action correction during the policy's own generation.

4.3. Restoring Shared Components Corrects Action Commands

Restoring shared response components inside $\pi_{0.5}$ corrects its initial action commands without retraining. At L9, the prespecified restoration site, we replace the components along the 16 shared directions with values obtained from a clean observation of the same physical state. At each planning query, the clean view is rendered at the intervention run's own current state. Clean and perturbed hidden states, H^c and H^p, are then computed using the same current action-plan sample and generation time:

$$H' = H^p - (H^p - H^c)BB^\top. \tag{4}$$

The projection replaces the coordinates along B while preserving the orthogonal components. Generation continues with the perturbed visual context. We apply this restoration at all ten generation steps of every planning query.

A separate confirmation study uses 216 paired cases across 31 tasks under Camera and Sensor perturbations, disjoint from nine development tasks. Task success is the prespecified primary endpoint; the key secondary endpoint is the Euclidean error over the first five translation commands issued at the initial query, measured against the paired unedited clean run. The unedited perturbed policy provides the correction baseline. Clean- and random-basis restoration test whether response directions give greater correction than other rank-16 choices; full L9 replacement provides a reference for restoring all 1024 channels. Each assigned basis excludes the test task group and perturbation family from fitting. Rank-16 restorations share a schedule. Each basis replaces its actual coordinate difference from the clean state, so the write magnitudes differ across bases (Supplement, Sec. S13).

Response restoration reduces the task-averaged command error from 0.409 to 0.312 for Camera and from 0.209 to

0.199 for Sensor, in control-interface units (Fig. 4). It also improves on clean- and random-direction restoration: all six simultaneous 95% intervals for error reduction against these controls and the unedited policy are positive. Full-state restoration gives errors of 0.297 and 0.194, respectively. Restoring the 16 shared directions achieves approximately 86.9% (Camera) and 70.7% (Sensor) of the mean command-error reduction obtained by replacing all 1024 channels (Supplement, Tables S32–S33).

To compare directions at a common target edit size, separate development controls rescale their projected edits at each plan position. In those saved-state generations, response directions give greater mean correction of the first five planned translations than clean directions for Camera, while clean directions give greater correction for Sensor (Supplement, Sec. S13.5, Table S34).

Observed task success rises from 73.15% to 75.00% for Camera and from 83.33% to 86.67% for Sensor, toward the clean baselines reported in Fig. 4. The figure places these primary outcomes beside command correction and gives the paired success gains with simultaneous 95% intervals over six success comparisons; the two displayed intervals include zero. With same-state clean observations as references, restoring shared components corrects the policy's initial action commands without retraining. This establishes a concrete use of the shared response structure: an internal correction that can be tested together with its effects on task completion.

5. From Action Commands to Motion and Task Outcomes

Section 4 identifies shared responses and tests their restoration during action generation. We now examine what action differences reveal about robot motion and task completion. The independent LIBERO-Plus execution study compares unedited $\pi_{0.5}$ and π_0 policies with and without visual perturbations, matching the initial physical state and policy-noise input (Sec. 3.2).

5.1. Initial Action Changes Track Early Motion

Initial translation-command differences closely track the early separation of the robot's end effector. We quantify the initial action change by the Euclidean norm of the difference between the first five issued translation commands, comprising 15 coordinates. We measure the distance between clean and perturbed end-effector positions after 5 and 60 issued controls. These are executed robot actions, distinct from the updates within one action-generation process (Sec. 3.1). The first five controls execute the first five commands from the initial plan; by control 60, further observations and repeated replanning have entered the execution. On the full 40-task panel, we compare the same initial command differences with motion at both times, using records available at both steps.

The task-weighted rank correlation falls from 0.953 at control 5 to 0.474 at control 60 in $\pi_{0.5}$, and from 0.974 to 0.409 in π_0. The early-minus-late differences are 0.479 $[0.368, 0.593]$ and 0.566 $[0.463, 0.671]$, respectively (95% task-bootstrap intervals; Supplement, Sec. S5.1). Thus, the initial action response is closely reflected in immediate motion, while its association with later motion is weaker after subsequent closed-loop interaction.

These action shifts also recur across the two recorded policy-noise inputs. For each visual setting, we compare the clean–perturbed differences in the first five issued commands under each noise input. An agreement score measures similarity in both direction and magnitude, with 1 denoting identical nonzero shifts. On the common-task panel, family-mean translation agreement ranges from 0.884 to 0.920 in $\pi_{0.5}$ and from 0.799 to 0.889 in π_0. The π_0 Sensor Noise value is 0.844 $[0.780, 0.896]$ (95% task-bootstrap interval). The same visual condition therefore induces similar initial translation changes under the two noise inputs (component results and valid supports: Supplement, Sec. S7).

5.2. Motion Magnitude and Paired Task Outcomes

Two clean runs can move differently because action generation uses different policy-noise inputs. We use this variation to judge the scale of the visual response: the task-equal mean clean–perturbed distance divided by the corresponding mean distance between clean runs. A ratio below 1 means that the visual response is smaller on average than this *clean-resampling reference*. Family-level comparisons use the 202 visual settings on the 28 common tasks, with two clean–perturbed comparisons per setting, one per policy-noise input. We compare this motion scale with each execution's final task outcome (Fig. 5).

Small early-motion differences accompany substantial success loss. Under Sensor Noise, π_0 loses 14.3 percentage points of task success, although its mean end-effector separation after five controls is 3.51 mm, below the 6.06 mm difference between clean runs. The motion ratio is 0.580 $[0.409, 0.791]$; the translation-command ratio is also below reference, at 0.545 $[0.395, 0.718]$. Both ratios use the same 55 settings and 28 tasks, with 95% intervals. The four-family-adjusted 98.75% interval for the success change is $[-28.6, -2.7]$ percentage points (Fig. 5a,b).

This contrast also holds within individual settings. Among the 110 pairs, 18 lose a clean success; 12 of these, spanning eight tasks, have early separation below their own setting's clean-resampling distance. These are unweighted pair counts; the motion and outcome rates above use equal task weighting. Both clean runs succeed in every setting containing a lost success. The small-motion contrast therefore

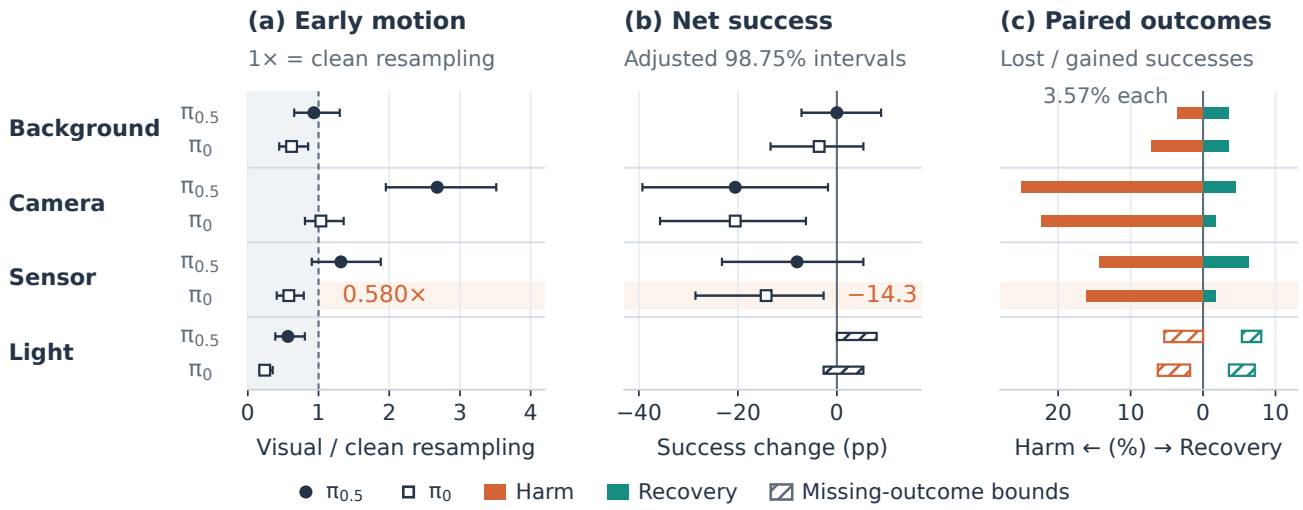


Figure 5. **Small motion can accompany success losses, and zero net change can hide outcome flips.** Read the highlighted π_0 Sensor Noise row across panels: (a) end-effector separation after five controls is below the variation between two clean runs with different policy noise (clean resampling), while (b) task success falls. (c) Lost clean successes (harm) extend left and gained successes (recovery) right; both lengths are positive event percentages. Each visual comparison matches the initial physical state and policy-noise input. The primary comparison uses 202 severity-labelled settings on 28 common tasks per policy. Motion ratios divide task-equal means, with pointwise 95% task intervals; shading marks values below 1. Success-change intervals are four-family-adjusted 98.75%. Hatching marks missing-outcome bounds, distinct from sampling intervals. Light motion covers 43/48 settings on 26 tasks ($\pi_{0.5}$) and 44/48 on 27 (π_0). Light outcome bounds retain all 48 planned settings. Outcome-flip comparisons and complete panels: Supplement, Secs. S6 and S18.

occurs among individual lost successes as well as in the family averages (post-hoc descriptive breakdown; Supplement, Table S10).

The same estimated success loss occurs at different motion scales. Under Camera perturbations, the estimated success drop is 20.5 percentage points in both policies, while the early-motion ratios are 2.68 in $\pi_{0.5}$ and 1.03 in π_0, each relative to its own clean-resampling reference (Fig. 5a,b). Together with the Sensor result, these comparisons show why early-motion magnitude and task outcomes need to be evaluated together.

Unchanged net success conceals lost and gained successes. We distinguish a clean success that becomes a failure (*harm*) from a clean failure that becomes a success (*recovery*). Both rates use all paired executions with the same task weights: their difference, recovery minus harm, gives the net success change, while their sum gives the fraction of outcomes that flip. Under $\pi_{0.5}$ Background, harm and recovery each occur at a weighted rate of 3.57%. They cancel in the net change, yet 7.14% of outcomes flip (Fig. 5c). The observed clean-resampling flip rate is 3.57% (complete comparisons: Supplement, Sec. S18). Thus, identical overall success rates can describe different sets of successful executions.

For VLA robustness studies, including evaluations of internal corrections, motion and paired task outcomes provide complementary evidence. Motion records how behavior changes; lost and recovered successes identify which executions are helped or harmed, even when the net success rate stays the same.

6. Conclusion

Our study identifies shared response directions that can be reused across visual conditions and used to correct action generation. In $\pi_{0.5}$, these directions describe responses on held-out tasks and perturbation families. Response reconstruction and fixed affine action prediction favor different sampled layers. Restoring the shared components using clean observations of the same physical state reduces the clean-reference error of the first five translation commands without retraining.

In independent executions, initial action differences closely track early motion. Yet substantial success losses occur even when early-motion differences lie below the clean-resampling reference, and unchanged net success can hide opposing outcome flips. Together, the findings give VLA robustness research a concrete starting point: shared components to test through internal correction, and measurements of actions, motion, and individual outcomes to determine what that correction achieves.

LLM Usage Statement

The authors conceived the research ideas and core experiments and made the final scientific decisions. LLM tools—ChatGPT and Codex (OpenAI), Claude (Anthropic), and Kimi (Moonshot AI)—assisted with implementing and debugging experimental and analysis code, refining experimental designs, and drafting and editing the manuscript and supplementary material. The authors checked the assisted work and take responsibility for the experiments, data, analyses, citations, and scientific conclusions.

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